(FP2) 3. Differentiation 专题作业1

NO. 1

It is given that

$$x = \sinh^{-1} t,\ y = \cos^{-1} t,$$

where $-1 < t < 1$.

a. By differentiating $\cos y$ with respect to t, show that $\frac{dy}{dt} = -\frac{1}{\sqrt{1-t^2}}$. [4]

b. Find $\frac{d^2y}{dx^2}$ in terms of t, simplifying your answer. [5]

NO. 2

The curve C has parametric equations

$$x = \frac{1}{2}t^2 - \ln t, \quad y = 2t + 1, \quad \text{for } \frac{1}{2} \leq t \leq 2.$$

a. Find the exact length of C. [5]

b. Find $\frac{d^2y}{dx^2}$ in terms of t, simplifying your answer. [4]

NO. 3

The variables x and y are such that y = −1 when x = 0 and

$$\left(x + \frac{\mathrm{d}y}{\mathrm{d}x}\right)^3 = y^2 + x.$$

i. Find the value of $\frac{\mathrm{d}y}{\mathrm{d}x}$ when x = 0. [1]

ii. Find also the value of $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ when x = 0. [4]

NO. 4

A curve C has equation $\cos y = x$, for $-\pi < x < \pi$.

i. Use implicit differentiation to show that $\frac{d^2y}{dx^2} = -\cot y\left(\frac{dy}{dx}\right)^2$. [4]

ii. Hence find the exact value of $\frac{d^2y}{dx^2}$ at the point $\left(\frac{1}{2}, \frac{1}{3}\pi\right)$ on C. [2]

NO. 5

A curve C has equation $x^3 - 3xy + y^2 = 4$. Find the value of $\frac{d^2y}{dx^2}$ at the point (0, 2) of C. [7]

NO. 6

A curve C has equation $\tan y = x$, for $x > 0$.

i. Use implicit differentiation to show that $\frac{d^2y}{dx^2} = -2x\left(\frac{dy}{dx}\right)^2$. [3]

ii. Hence find the value of $\frac{d^2y}{dx^2}$ at the point $(1, \frac{1}{4}\pi)$ on C. [2]

NO. 7

i. A curve C has equation $x^2 + 4xy - y^2 + 20 = 0$. Show that, at stationary points on C, $x = -2y$. [3]

ii. Find the coordinates of the stationary points on C, and determine their nature by considering the value of $\frac{d^2y}{dx^2}$ at the stationary points. [8]

NO. 8

It is given that

$$x = 1 + \frac{1}{t} \quad \text{and} \quad y = \cos^{-1} t \quad \text{for } 0 < t < 1.$$

a. Show that $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{t^2}{\sqrt{1-t^2}}$. [2]

b. Show that $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -t^a\left(1-t^2\right)^b\left(2-t^2\right)$, where a and b are constants to be determined. [4]

NO. 9

It is given that

$$x = \sin^{-1} t \quad \text{and} \quad y = t\cos^{-1} t, \qquad \text{for } 0 \leqslant t < 1.$$

a. Show that $\frac{dy}{dx} = -t + \sqrt{1-t^2}\cos^{-1} t$. [3]

b. Find $\frac{d^2y}{dx^2}$ in terms of t. [4]

NO. 10

It is given that

$$x = 1 + \frac{1}{t} \qquad \text{and} \qquad y = te^t.$$

a. Show that $\frac{dy}{dx} = -e^t(t^3 + t^2)$. [3]

b. Find $\frac{d^2y}{dx^2}$ in terms of t. [4]